

INTERNATIONAL FALLS, MN, USA

WMO: 727470

Lat: 48.561N

Lon: 93.398W

Elev: 361

StdP: 97.07

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14918

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.4	(c) -29.3	(d) -34.6	(e) 0.1	(f) -31.4	(g) -32.2	(h) 0.2	(i) -28.6	(j) 9.3	(k) -10.9	(l) 8.6	(m) -12.1	(n) 1.6	(o) 290	(p) 0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	29.5	21.0	27.9	19.8	26.4	18.9	22.7	27.4	21.4	26.1	20.3	24.5	4.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.3	25.6	19.8	15.1	24.0	18.6	14.1	22.8	68.6	27.2	63.9	26.2	59.9	24.4	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.9	8.0	7.3		-37.1	32.2	3.9	1.7	-39.9	33.4	-42.2	34.4	-44.4	35.4	-47.2	36.6	
				DB												
				WB	-35.4	24.6	2.7	1.1	-37.4	25.4	-38.9	26.1	-40.4	26.7	-42.3	27.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-14.7	-12.2	-4.4	3.7	10.7	16.3	18.6	17.3	12.9	5.7	-2.8	-11.0
	DBStd	13.02	8.35	7.82	7.57	5.41	4.79	3.87	3.19	3.36	4.63	4.86	6.35	7.64
	HDD10.0	3292	767	621	447	198	49	2	0	1	22	152	384	650
	HDD18.3	5547	1025	854	704	439	240	83	36	59	172	393	633	908
	CDD10.0	895	0	0	1	8	71	191	268	228	109	18	1	0
	CDD18.3	108	0	0	0	0	4	21	46	28	9	1	0	0
	CDH23.3	1296	0	0	1	4	84	267	504	331	99	6	0	0
	CDH26.7	282	0	0	0	1	18	62	114	69	17	1	0	0
(14) Wind	WSAvg	3.3	3.3	3.4	3.6	3.7	3.7	3.2	2.9	2.8	3.1	3.5	3.5	3.2
(15) Precipitation	PrecAvg	627	21	16	26	41	71	99	95	74	77	52	33	23
	PrecMax	832	93	40	95	91	169	260	242	169	187	123	89	55
	PrecMin	428	0	1	3	2	5	18	27	15	7	4	0	5
	PrecStd	95	17	10	19	22	39	48	51	33	44	31	21	12
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.4	7.4	17.7	22.9	29.1	31.7	31.3	30.6	28.8	23.8	15.5	5.4
		MCWB	1.6	4.1	10.7	13.5	17.4	22.6	22.5	22.1	20.3	15.6	10.2	2.3
	2%	DB	1.3	4.3	11.6	19.4	25.8	28.2	29.5	28.6	26.2	19.9	11.2	2.5
		MCWB	0.0	1.9	7.1	10.3	15.8	19.6	21.4	20.8	18.9	13.5	7.8	0.9
	5%	DB	-0.7	1.8	8.3	16.4	22.8	26.3	27.8	26.9	23.8	16.8	8.2	0.9
		MCWB	-1.8	0.2	4.2	8.6	13.9	18.2	20.3	19.7	17.7	12.2	5.4	-0.1
	10%	DB	-2.7	-0.3	5.5	13.5	20.3	24.4	26.3	25.1	21.5	13.8	5.7	-0.6
		MCWB	-3.6	-1.8	2.4	7.2	12.9	17.4	19.4	18.6	16.7	9.9	3.5	-1.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	4.5	12.3	14.3	19.7	23.5	24.2	23.7	22.0	17.0	11.3	2.8
		MCDB	3.7	6.5	14.7	20.1	25.2	29.4	29.6	28.7	27.1	20.7	14.2	4.4
	2%	WB	0.3	1.9	7.4	11.2	17.5	21.4	22.8	22.2	20.2	14.7	7.9	1.2
		MCDB	1.4	3.6	11.6	17.6	22.6	26.7	27.2	27.1	24.1	19.4	10.8	2.2
	5%	WB	-1.7	0.4	4.7	9.3	15.5	19.9	21.6	20.8	18.7	12.4	5.7	0.0
		MCDB	-0.7	1.8	7.8	15.4	20.8	23.9	26.1	25.1	22.7	16.1	7.9	1.0
	10%	WB	-3.6	-1.8	2.6	7.6	13.9	18.5	20.4	19.7	17.1	10.3	3.6	-1.6
		MCDB	-2.6	-0.3	5.2	13.3	19.1	22.7	24.7	23.7	20.6	13.2	5.5	-0.6
(35) Mean Daily Temperature Range	MDBR		10.7	12.1	11.8	12.1	13.1	12.6	13.0	13.2	12.3	9.9	8.2	9.0
	5% DB	MCDBR	9.9	11.6	13.6	17.8	17.9	15.4	14.9	15.1	15.4	14.9	11.8	9.2
		MCWBR	8.7	9.3	9.1	10.5	9.2	8.1	7.8	8.1	8.9	9.3	8.3	7.9
	5% WB	MCDBR	9.3	10.3	12.5	15.8	14.4	12.6	12.7	13.0	12.6	13.3	10.8	8.7
		MCWBR	8.2	8.3	8.8	9.8	8.2	7.3	7.5	7.7	7.9	8.7	8.0	7.7
(40) Clear-Sky Solar Irradiance	taub		0.250	0.268	0.308	0.339	0.355	0.373	0.384	0.372	0.354	0.305	0.284	0.254
	taud		2.365	2.323	2.364	2.403	2.413	2.385	2.375	2.440	2.462	2.585	2.505	2.401
	Ebn at Noon		837	897	909	911	906	888	871	868	850	853	793	783
	Edh at Noon		66	87	101	108	111	116	115	103	92	69	59	57
(44) All-Sky Solar Radiation	RadAvg		1.30	2.50	3.74	4.78	5.26	5.57	5.88	4.98	3.67	2.12	1.17	0.92
	RadStd		0.17	0.26	0.34	0.43	0.35	0.44	0.34	0.34	0.31	0.29	0.15	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page